UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 004 · Particular integral Type 2 (sin ax, cos ax) · compiled 16 October 2026

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1 Short notes

1 · The Type 2 rule: replace D^2 by $-a^2$

Type 2: $X = \sin(ax+b)$ or $\cos(ax+b)$. PI $= \frac{X}{f(-a^2)}$: replace every D^2 by $-a^2$ (minus a squared), $D^3 = D^2D \to -a^2D$. If a lone D remains, $\frac{1}{p+qD} = \frac{p-qD}{p^2-q^2D^2}$, replace D^2 again, and let $p-qD$ differentiate X. Example: $\frac{\cos(2x+3)}{D^3+1} = \frac{\cos(2x+3)-8\sin(2x+3)}{65}$; CW P13: $\frac{\sin 3x}{D^3-D^2+4D-4} = \frac{3\cos 3x+\sin 3x}{50}$.

2 · Case of failure: $f(-a^2) = 0$

Failure: $f(-a^2) = 0 \iff D^2+a^2$ divides $f(D)$ ($\pm ai$ in the CF). Then PI $= x\frac{X}{f'(-a^2)}$, f' w.r.t. D; if f' also fails, $x^2\frac{X}{f''(-a^2)}$. Standard: $\frac{\sin ax}{D^2+a^2} = -\frac{x\cos ax}{2a}$, $\frac{\cos ax}{D^2+a^2} = \frac{x\sin ax}{2a}$; $\frac{\cos\sqrt{3}x}{D^2+3} = \frac{x\sin\sqrt{3}x}{2\sqrt{3}}$. ESA ×3: $(D-2)^2y = 8(e^{2x}+\sin 2x)$: $y = (c_1+c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x$.

3 · Products, powers and sums: reducing to Type 2

Reduce first: $\sin A\cos B = \frac{1}{2}[\sin(A+B)+\sin(A-B)]$; $\sin^2 = \frac{1-\cos 2ax}{2}$, $\cos^2 = \frac{1+\cos 2ax}{2}$; $\sin^3 A = \frac{3\sin A-\sin 3A}{4}$. The constant is Type 1 ($D \to 0$; fails if D divides f). Add the separate PIs. $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$: PI $= \frac{x^2e^x}{4} + \frac{1}{2} + \frac{x\cos x}{8}$ (slide prints $\frac{1}{4}$). $\frac{\sin^2 x}{D^3-2D^2+D} = \frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$ (slide drops a 2).

2 Keyword index

19 terms; the number in brackets is the topic.

$\frac{1}{D}$ (2). Integration, no constant.

$D^2 + a^2$ **factor** (2). Why $f(-a^2) = 0$; $\pm ai$ in the CF.

$D^2 \to -a^2$ (1). The Type 2 replacement; minus a squared.

$f(-a^2)$ (1). Denominator after the replacement.

Case of failure (2). $f(-a^2) = 0$.

Constant term (3). ke^{0x}: Type 1 with $D \to 0$.

Factor by factor (2). Apply the non-failing factor first.

Half-angle identity (3). $\sin^2 = \frac{1-\cos 2ax}{2}$.

Multiply by x (2). $x\frac{X}{f'(-a^2)}$, f' w.r.t. D.

Odd powers (1). $D^3 = D^2 \cdot D$, $D^5 = (D^2)^2 D$.

Operator on the left (1). The function stays to the right of every operator.

Phase shift (1). PI $= A\cos + B\sin$ at the same frequency.

Rationalise (1). Multiply by the conjugate $p - qD$.

Resonance (2). PI $\propto x\sin$, $x\cos$: grows.

Standard result (2). $\frac{\cos ax}{D^2+a^2} = \frac{x\sin ax}{2a}$.

Superposition (3). PI $= \mathrm{PI}_1 + \mathrm{PI}_2 + \ldots$

Transformation formula (3). Product to sum.

Triple-angle identity (3). $\sin^3 A = \frac{3\sin A - \sin 3A}{4}$.

Type 2 (1). $X = \sin(ax+b)$ or $\cos(ax+b)$.

3 Rapid recall

1	Type 2 rule?	$\frac{\sin(ax+b)}{f(D^2)} = \frac{\sin(ax+b)}{f(-a^2)}$, $f(-a^2) \neq 0$.
2	For $\cos 2x$, D^2 becomes?	-4 (not $+4$).
3	D^3 for $\sin 3x$?	$D^2 \cdot D \to -9D$.
4	How to remove a lone D in $\frac{1}{p+qD}$?	Multiply by $p - qD$: $\frac{p-qD}{p^2-q^2D^2}$.
5	$\frac{\cos(2x+3)}{D^3+1}$?	$\frac{\cos(2x+3)-8\sin(2x+3)}{65}$.
6	$\frac{\sin 3x}{D^3-D^2+4D-4}$ (CW P13)?	$\frac{3\cos 3x+\sin 3x}{50}$.
7	When does $f(-a^2) = 0$?	When $D^2 + a^2$ is a factor of $f(D)$ ($\pm ai$ AE roots).
8	PI on failure?	$x\frac{X}{f'(-a^2)}$, f' w.r.t. D.
9	$\frac{\cos\sqrt{3}x}{D^2+3}$?	$\frac{x\sin\sqrt{3}x}{2\sqrt{3}}$.
10	$\frac{\sin ax}{D^2+a^2}$?	$-\frac{x\cos ax}{2a}$.
11	$\frac{\cos ax}{D^2+a^2}$?	$\frac{x\sin ax}{2a}$.
12	Solve $(D-2)^2y = 8(e^{2x} + \sin 2x)$?	$y = (c_1 + c_2x)e^{2x} + 4x^2e^{2x} + \cos 2x$.
13	$\frac{8\sin 2x}{(D-2)^2}$?	$\frac{8\sin 2x}{-4D} = \cos 2x$.
14	$\sin^2 x$ rewritten?	$\frac{1-\cos 2x}{2}$.
15	Constant part of $\frac{\cos^2(x/2)}{(D-1)^2(D^2+1)}$?	$\frac{1}{2}$ (slide 004·14 prints $\frac{1}{4}$).
16	PI of $(D-1)^2(D^2+1)y = e^x + \cos^2\frac{x}{2}$?	$\frac{x^2e^x}{4} + \frac{1}{2} + \frac{x\cos x}{8}$.
17	PI of $(D^3 - 2D^2 + D)y = \sin^2 x$?	$\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$ (slide drops a 2).
18	$\sin^3 x$ rewritten?	$\frac{3\sin x-\sin 3x}{4}$.

Mnemonic (study aid). *"Square goes negative, odd D gets a partner."* $D^2 \to -a^2$; a lone D is rationalised with its conjugate (study aid).

4 Last 10 minutes before the exam

Top facts

1. Type 2: $\frac{X}{f(D^2)} = \frac{X}{f(-a^2)}$; replace D^2 by $-a^2$ (minus a squared), $D^3 = D^2 D$.
2. Lone D left: multiply by the conjugate, $\frac{1}{p+qD} = \frac{p-qD}{p^2+q^2a^2}$; the numerator differentiates X.
3. Failure $f(-a^2) = 0$ ($D^2 + a^2$ divides f): PI $= x\frac{X}{f'(-a^2)}$; $\frac{\cos ax}{D^2+a^2} = \frac{x \sin ax}{2a}$, $\frac{\sin ax}{D^2+a^2} = -\frac{x \cos ax}{2a}$.
4. $\frac{1}{D}$ = integrate (no constant); the x stays outside.
5. ESA ×3: $(D-2)^2 y = 8(e^{2x} + \sin 2x) \Rightarrow y = (c_1 + c_2 x)e^{2x} + 4x^2 e^{2x} + \cos 2x$.
6. Products → sums; $\sin^2, \cos^2 \to \frac{1}{2} \mp \frac{1}{2}\cos 2ax$ (the constant is Type 1, $D \to 0$); $\sin^3 A = \frac{3\sin A - \sin 3A}{4}$.
7. Slide slips: 004·14 constant is $\frac{1}{2}$ (not $\frac{1}{4}$); 004·17 PI is $\frac{x}{2} - \frac{\cos 2x}{25} + \frac{3\sin 2x}{100}$.

Top formulas

E1.1 · Type 2 rule

ESA ★★★

$$\frac{1}{f(D^2)}\sin(ax+b) = \frac{\sin(ax+b)}{f(-a^2)}$$
$$\frac{1}{f(D^2)}\cos(ax+b) = \frac{\cos(ax+b)}{f(-a^2)}$$

GIVES The PI of a sine or cosine.

APPLIES $f(-a^2) \neq 0$, i.e. $\pm ai$ not roots of the AE.

NOTE Replace D^2 by $-a^2$, not by $(-a)^2$; b plays no part.

S004·4 · S004·15 · T004 00:00:58 · T004b 00:06:32

E1.3 · Rationalising a linear denominator

ESA ★★★

$$\frac{1}{p+qD}X = \frac{(p-qD)}{p^2-q^2D^2}X = \frac{(p-qD)X}{p^2+q^2a^2}$$

GIVES A PI with no D left in the denominator.

APPLIES After E1.2 a single D remains; $X = \sin(ax+b)$ or $\cos(ax+b)$.

NOTE $p - qD$ then acts on X: $D\sin(ax+b) = a\cos(ax+b)$, $D\cos(ax+b) = -a\sin(ax+b)$.

S004·9 · S004·17 · T004 00:12:01 · T004b 00:11:14

E2.1 · First case of failure

ESA ★★★

$$f(-a^2)=0: \quad \frac{1}{f(D^2)}\sin(ax+b)=x\,\frac{\sin(ax+b)}{f'(-a^2)} \quad \text{(same for cos)}$$

GIVES The PI when the frequency is in the CF.

APPLIES $f(-a^2)=0$, $f'(-a^2)\neq 0$.

NOTE f' = derivative w.r.t. D; it usually leaves a D (e.g. $2D$), which is then integration $\frac{1}{D}$.

S004·5 · S004·15 · S004·16 · T004 00:03:07 · T004b 00:07:32

E2.2 · Standard results for D^2+a^2

ESA ★★★

$$\frac{1}{D^2+a^2}\sin ax=-\frac{x\cos ax}{2a}, \qquad \frac{1}{D^2+a^2}\cos ax=\frac{x\sin ax}{2a}$$

GIVES The failure PI in one line.

APPLIES $f(D)=D^2+a^2$ (or this factor, with the rest applied first).

NOTE From E2.1: $\frac{x}{2D}\sin ax=\frac{x}{2}\int \sin ax\,dx$.

ADDED FOR CLARITY S004·16 · T004 00:23:17

E3.2 · Squares

ESA ★★★

$$\sin^2 ax=\frac{1-\cos 2ax}{2}, \qquad \cos^2 ax=\frac{1+\cos 2ax}{2}$$

GIVES A constant (Type 1 with $a=0$) plus a Type 2 term at frequency $2a$.

APPLIES $X=\sin^2 ax$ or $\cos^2 ax$.

NOTE $\cos^2\frac{x}{2}=\frac{1+\cos x}{2}$, $\sin^2\frac{x}{2}=\frac{1-\cos x}{2}$.

S004·7 · S004·13 · S004·17 · T004 00:06:23 · T004 00:20:36 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E3.4 · Superposition

ESA ★★★

$$\frac{1}{f(D)}(X_1+X_2+\dots)=\frac{1}{f(D)}X_1+\frac{1}{f(D)}X_2+\dots=\mathrm{PI}_1+\mathrm{PI}_2+\dots$$

GIVES The PI of a sum of terms of different types.

APPLIES X is a sum, e.g. $e^x+\cos^2\frac{x}{2}$.

NOTE Each term chooses its own rule (Type 1, Type 2, failure).

S004·11 · S004·14 · T004 00:17:10 · ESA Dec-23 3a